# MUSIC & MENTAL HEALTH

## MID-TERM PROJECT

Presented by:
An Nguyễn & Jasmine Hardy

**PHY-231: Think Like a Data Scientist**

# QUESTION 1

## Are types of mental health issues and the choice of genre associated?

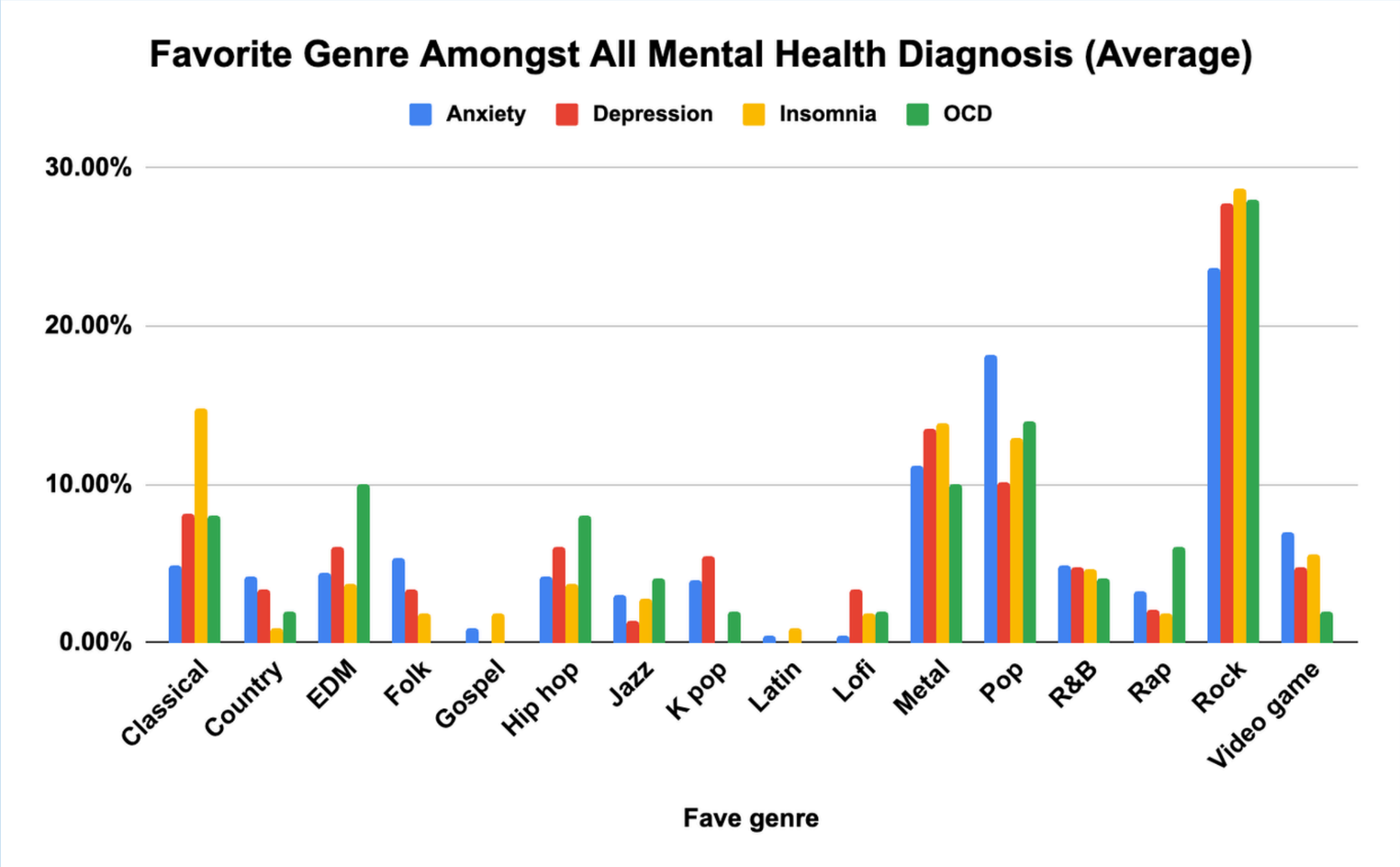



Figure 1. The bar chart displays the comparison of the favorite genres, on average, amongst each mental health diagnosis. There is no major difference between the diagnoses.

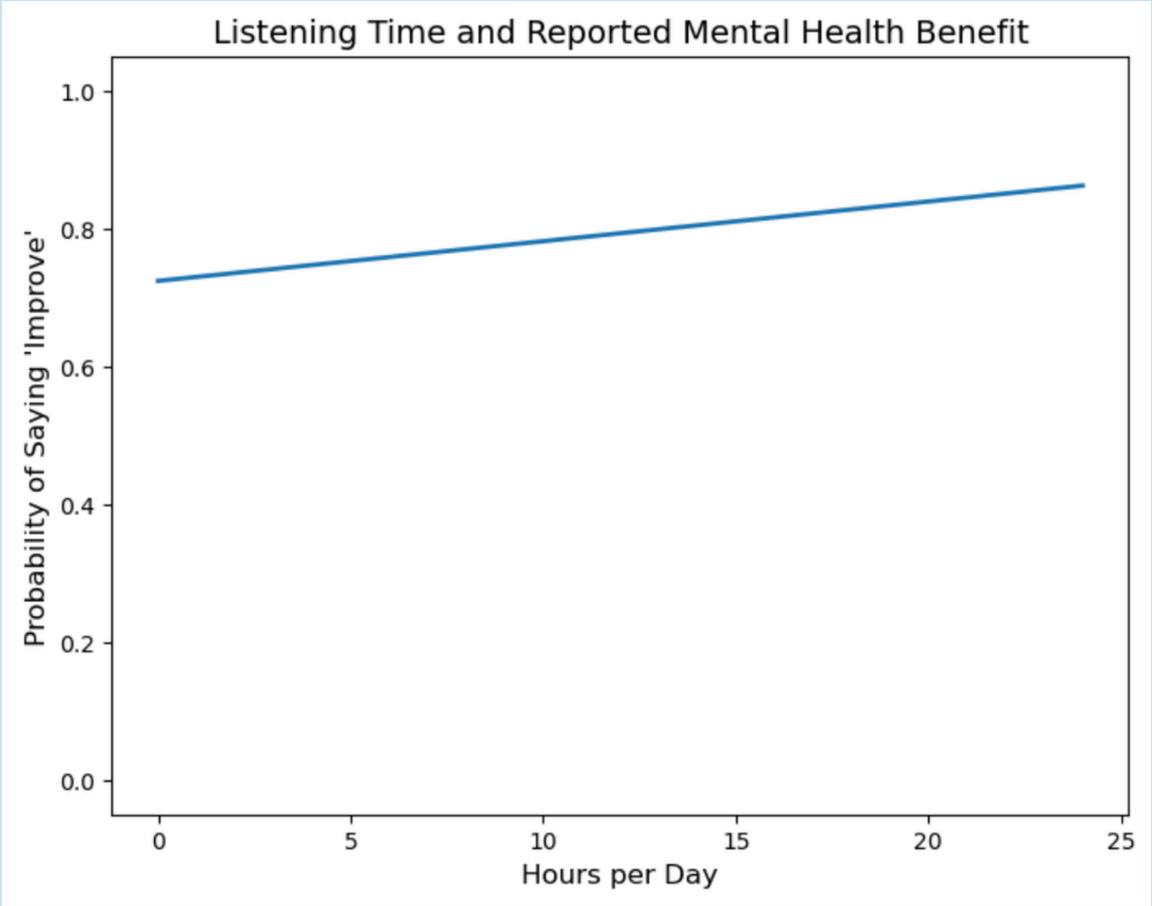
**Result: The bar chart suggests there are no major differences between mental health diagnoses and the choice of genre.**

However, there are some interesting observations

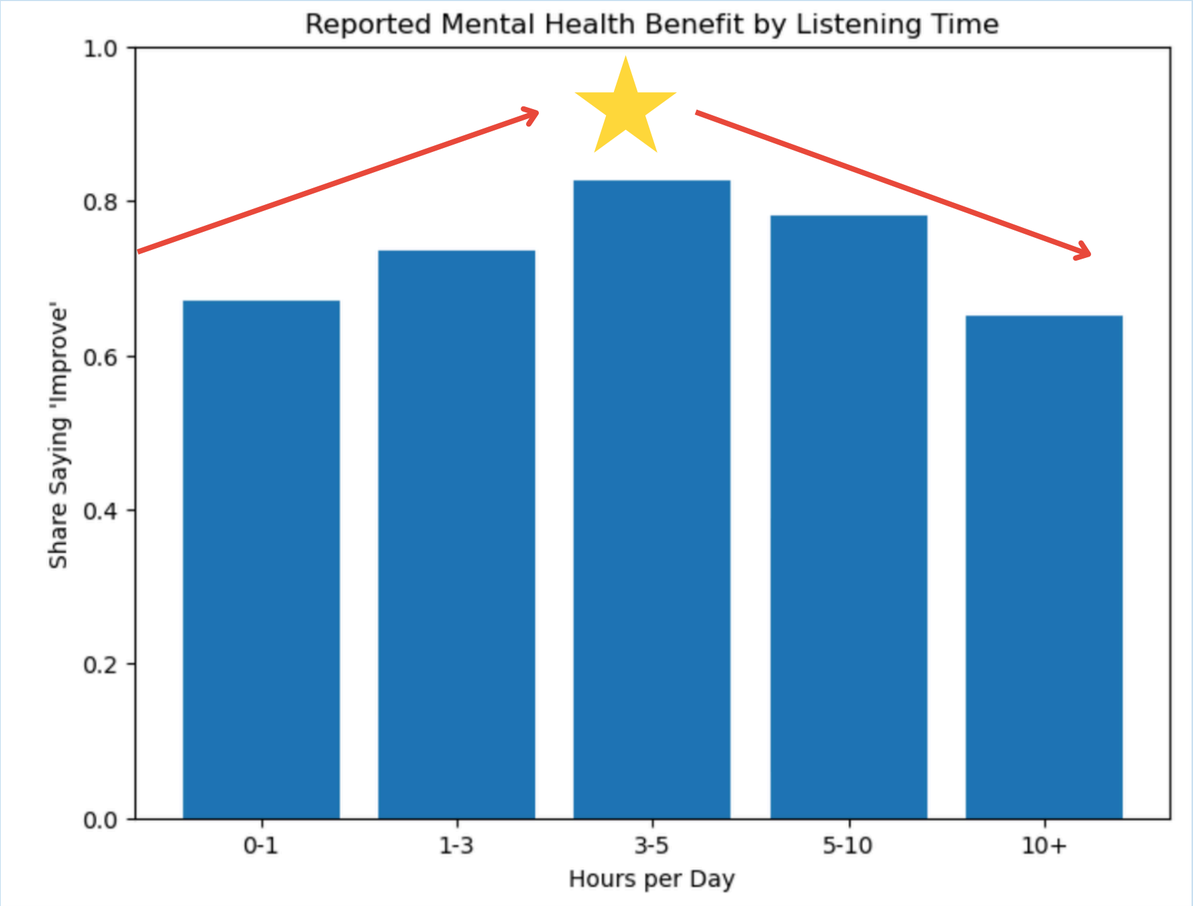
- Top 3 favorite genres amongst all diagnoses are Rock, Pop, and Metal.
- Insomnia respondents have a higher average of classical music listeners than other diagnoses.
- OCD respondents have a higher average of EDM listeners than other diagnoses

# QUESTION 2

**Is there a relationship between hours spent listening and reported mental health benefits?**




**Is more listening always better, or is there a point of diminishing returns?**



# QUESTION 3

**Do people who say music improves their mental health actually have higher baseline anxiety and depression? In other words, are the people who benefit most the ones who started from a harder place?**

**Regression allows us to:**

- Compare two groups and tell us how big the difference is between them
- Tell us whether the difference is likely real or just due to random chances with the p-value.

**Focused on two groups:**

- Improve
- No Effect

**Ran two simple models:**

Anxiety = $\alpha_0 + \alpha_1$(Improve)

Depression = $\beta_0 + \beta_1$(Improve)

***The coefficient on "Improve" tells us how much higher (or lower) baseline anxiety or depression is for those who report benefiting from music.***

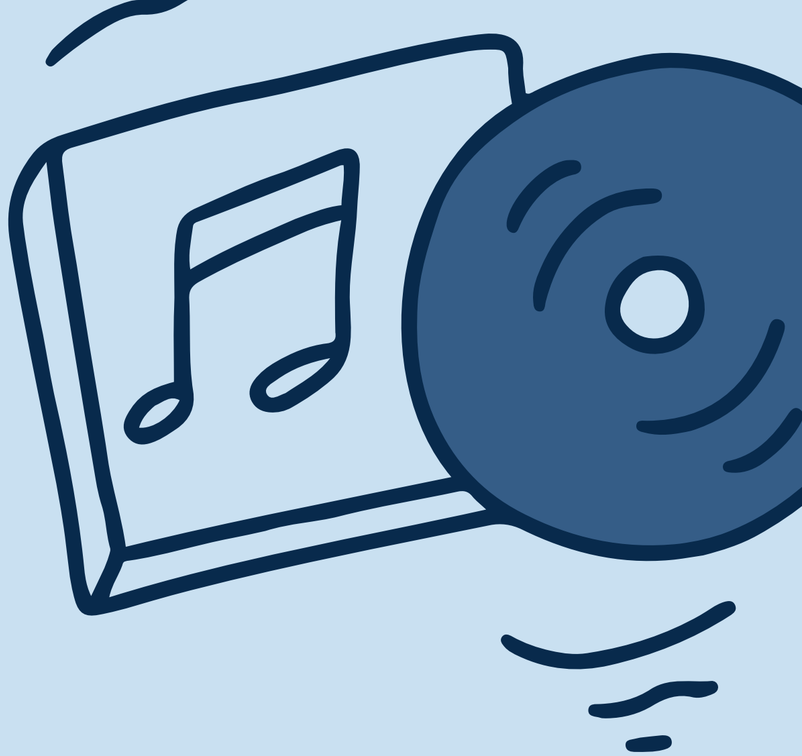
# STATISTICAL REGRESSION RESULTS

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                Anxiety   R-squared:                       0.015
Model:                            OLS   Adj. R-squared:                  0.013
Method:                 Least Squares   F-statistic:                     10.79
Date:                Wed, 18 Feb 2026   Prob (F-statistic):            0.00107
Time:                        19:56:35   Log-Likelihood:                -1773.4
No. Observations:                 728   AIC:                             3551.
Df Residuals:                     726   BIC:                             3560.
Df Model:                           1
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept          5.2742      0.203     25.978      0.000       4.876       5.673
Improve_dummy      0.7729      0.235      3.285      0.001       0.311       1.235
==============================================================================
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              Depression   R-squared:                       0.001
Model:                             OLS   Adj. R-squared:                 -0.000
Method:                  Least Squares   F-statistic:                    0.7815
Date:                 Wed, 18 Feb 2026   Prob (F-statistic):              0.377
Time:                         19:49:00   Log-Likelihood:                -1837.7
No. Observations:                  728   AIC:                             3679.
Df Residuals:                      726   BIC:                             3689.
Df Model:                            1
Covariance Type:             nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept          4.6344      0.222     20.897      0.000       4.199       5.070
Improve_dummy      0.2272      0.257      0.884      0.377      -0.277       0.732
==============================================================================
```

- **Anxiety Analysis:**
  - Regression Coefficient (α) = 0.77.
  - P-value = 0.001 (Statistically Significant).
  - ***Result: Those who say music improves their mental health report anxiety levels 0.77 points higher on average than those reporting no effect.***

- **Depression Analysis:**
  - Regression Coefficient (β) = 0.23.
  - P-value = 0.377 (Not Statistically Significant).
  - ***Result: There is no strong statistical evidence that the "Improve" group has a higher baseline for depression.***



# THANK YOU!